TitrateRL: A Simulation-Based Reinforcement Learning Approach for Personalized Basal Insulin Titration in Type 2 Diabetes

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***Abstract*—TitrateRL proposes an open, simulation-based research framework for personalized basal-insulin titration in Type 2 diabetes (T2D). A published T2D glucose–insulin model is extended with long-acting glargine absorption, meal variability and sensor noise, calibrated against a landmark titration trial, and exposed as a Gymnasium environment with 150 heterogeneous virtual patients. Proximal Policy Optimization (PPO) agents with a rule-based safety shield will be compared with three clinical titration algorithms over 24-week simulated courses. The implemented simulator passes integrator, equilibrium, cross-implementation and glargine-profile checks, and one 168-day episode runs in about 0.15 s; calibration against the Treat-to-Target trial is still in progress. Evaluation of reward design, comparison with clinical algorithms and analysis of personalization on held-out virtual patients is planned. All results are obtained in silico and are not intended for clinical use.**

***Index Terms*—reinforcement learning, Type 2 diabetes, basal insulin titration, in silico simulation, safe reinforcement learning, Gymnasium**

I. Introduction

Type 2 diabetes (T2D) affects most of the 589 million adults living with diabetes [1]. When non-insulin agents no longer reach the glycemic target, a once-daily basal insulin such as glargine is added and its dose is titrated, for example by +2 U every three days, until the fasting plasma glucose (FPG) reaches 80–130 mg/dL [2], [3]. Clinical titration algorithms apply the same thresholds to every patient, and in routine care up-titration is slow: 81% of patients remained above their HbA1c target six months after starting basal insulin [4]. This project investigates whether a reinforcement learning (RL) policy, trained entirely in simulation, can titrate basal insulin more effectively than clinical rules without increasing hypoglycemia, and builds the open infrastructure needed to answer this question reproducibly.

A. Background and Related Work

The treat-to-target, AT.LANTUS and INSIGHT trials established weekly, three-day and daily fasting-glucose titration rules [5]–[7]. The meal simulation model of Dalla Man *et al.* [8] is the basis of the UVA/Padova simulator, accepted for preclinical testing of insulin-dosing algorithms [9]; its open re-implementation `simglucose` contains only Type 1 patients [10]. Bonet *et al.* showed in silico that CGM-based rules reach the optimal T2D dose faster than guideline titration [11]. RL improves glucose control in Type 1 simulations but can fail catastrophically when applied naively [12]; 24 of 29 RL glucose-control studies address Type 1 diabetes [13]. RL for T2D has used non-public hospital records [14] or a commercial simulator [15], and a recent open safe-RL testbed keeps the basal dose fixed [16]. Table I summarizes the design choices these studies motivate.

B. Problem Statements

P1: Formulation. Daily T2D basal titration has not been posed as an RL problem in an open environment, and it is unclear how clinical objectives with strongly asymmetric risks should be encoded in a scalar reward.

P2: Performance. Clinical algorithms are population-level and use only fasting glucose. Whether a policy learned in simulation keeps glucose in range more effectively without increasing hypoglycemia is unknown for T2D.

P3: Heterogeneity. Insulin sensitivity varies widely between patients and is not observed directly [17]. How the benefit and the dosing behavior of a learned policy depend on insulin sensitivity has not been characterized.

C. Research Questions and Specific Objectives

Each research question is paired with a specific, measurable objective (SO) that fixes the method, the metrics, the success criterion and the deadline in advance.

1) **RQ1.** How does the reward formulation (R1 time in range, R2 risk index, R3 time in range with hypoglycemia penalties) affect time in range (TIR) and time below range (TBR) of PPO titration policies?
   SO1 (Weeks 8–10). Train PPO under R1, R2 and R3 (five seeds each, shield off) in the environment validated at milestone M1, and select the reward $R^{\star}$ with the highest median validation TIR among the rewards for which $\mathrm{TBR}_{<70} < 4\%$ and $\mathrm{TBR}_{<54} < 1\%$ hold in at least 90% of validation patients. Evaluate the three configurations on the 30-patient test cohort and report the median and 95% bootstrap CI of TIR, $\mathrm{TBR}_{<70}$ and $\mathrm{TBR}_{<54}$. *Success criterion (H1):* R3 has a lower $\mathrm{TBR}_{<70}$ than R1 (one-sided

TABLE I
RELATED STUDIES AND IMPLICATIONS FOR TITRATERL

Study	Method or setting	Implication for TitrateRL
Dalla Man *et al.* [8]	Meal simulation model with published T2D parameters	Implement the T2D model openly in Python and add subcutaneous basal insulin.
Kovatchev *et al.* [9]	In silico preclinical trials (UVA/Padova, T1D)	Verify and calibrate the simulator before any policy is trained.
Xie, `simglucose` [10]	Open Python T1D simulator with a Gym interface	Reuse the shared subsystems and test against them (cross-implementation).
Riddle, Davies, Gerstein [5]–[7]	Clinical fasting-glucose titration algorithms	Implement them as baselines B1–B3 and calibrate against Treat-to-Target.
Bonet *et al.* [11]	CGM-based titration rules, 300 virtual T2D subjects	Use CGM features and compare titration speed (time-to-target).
Fox *et al.* [12]	Deep RL for closed-loop control in T1D	Use validation-based checkpoint selection and multi-seed reporting.
Wang *et al.* [14]	Model-based RL from hospital records (T2D)	Provide an open benchmark that does not depend on non-public data.
Panagiotou *et al.* [15]	Actor-critic advisor in a commercial T2D simulator	Keep the simulator, cohort and code fully open and reproducible.
Kwon *et al.* [16]	Safe-RL testbed; safety fails on unseen patients	Evaluate on held-out patients and add a rule-based safety shield.

paired Wilcoxon test, $\alpha = 0.05$) with a median TIR at most 5 pp lower.

2) **RQ2.** Does a shielded PPO policy achieve a higher maintenance-phase TIR than the AT.LANTUS, Treat-to-Target and INSIGHT algorithms on held-out patients while remaining non-inferior in $\text{TBR}_{<70}$?
   SO2 (Week 11). Evaluate the primary policy, PPO with $R^\star$ and the safety shield (five seeds), against B1–B3 on the 30 test patients with ten 24-week scenarios each. The primary efficacy endpoint is TIR in weeks 21–24 and the primary safety endpoint is $\text{TBR}_{<70}$ over 24 weeks. *Success criterion (H2), per algorithm:* (i) the Holm-adjusted one-sided paired Wilcoxon test of the TIR difference is significant ($\alpha = 0.05$); (ii) the Hodges–Lehmann TIR gain is at least 5 pp; and (iii) the upper limit of the 95% bootstrap CI of the mean paired $\text{TBR}_{<70}$ difference is below 1 pp.
3) **RQ3.** How does the benefit vary with insulin sensitivity, and does the policy choose different dose changes for insulin-resistant and insulin-sensitive patients with similar fasting glucose?
   SO3 (Week 13). For the 30 test patients, quantify (i) Spearman's correlation between the insulin-sensitivity factor ρ_{SI} and the per-patient TIR difference to the best clinical algorithm; (ii) the difference between the lowest and the highest ρ_{SI} tertile in the mean dose change chosen at matched fasting glucose (100–130, 130–180 and >180 mg/dL); and (iii) the train–test generalization gap in TIR and $\text{TBR}_{<70}$, all with 95% bootstrap CIs. *Success criterion (H3):* a negative correlation (one-sided, $\alpha = 0.05$) and larger dose increases for the most insulin-resistant tertile (CI of the difference excludes zero).

D. General Objective

Design, implement and evaluate within the 14-week semester an open-source, simulation-based RL framework that learns daily basal-insulin titration policies for virtual patients with T2D and benchmarks them against clinical titration algorithms using internationally recommended glycemic metrics. The framework is delivered as an installable, validated environment with an evaluation pipeline and an interactive dashboard.

II. METHODOLOGY

A. Development Process and Project Scope

The team follows an iterative and incremental process inspired by Scrum, with two-week sprints, a prioritized backlog in GitHub Projects, weekly synchronization and supervisor reviews at milestones M1–M4. An item is done when its code is merged, its tests pass in continuous integration, it is documented, and any experiment it produced is logged with its configuration and commit hash.

Within this process, the simulator, the environment wrappers and the metric functions are built with *test-driven development* (TDD): each requirement is first written as a failing `pytest` test, and the code is written until the test passes. The test suite includes (i) an integrator test (the RK4 trajectory matches SciPy's `solve_ivp`); (ii) a cross-implementation test (with endogenous secretion switched off and `simglucose` parameters, the shared subsystems reproduce the `simglucose` right-hand side); (iii) equilibrium and periodic-steady-state tests; (iv) a dose–response test (fasting glucose is non-increasing in the dose); (v) tests that reproduce the published titration steps of B1–B3; and (vi) metric tests against hand-computed examples, rate-limit and shield tests, determinism tests and the Gymnasium API check. These tests also implement the simulator acceptance checks of gate G1.

GitHub Flow with a protected `main` branch and reviewed pull requests separates development from stable revisions; continuous integration runs linting and the test suite on Python 3.10–3.12. Experiments are version-controlled YAML files with fixed seeds, and models and result tables are stored separately from the source code. The project covers in silico research software only: real patient data, dosing recommendations for real people, Type 1 diabetes, bolus insulin and regulatory validation are outside its scope, and every output is labelled as a simulation result [18].

B. System Architecture and Deployment

TitrateRL follows a layered architecture in which each layer depends only on the layers below it: configuration and data, simulation core, environment, decision, experiment and evaluation, and presentation. Fig. 1 shows where each part runs and how artifacts move between the team's workstations, GitHub (repository, CI and releases), the training machine and the users of the package and the dashboard.

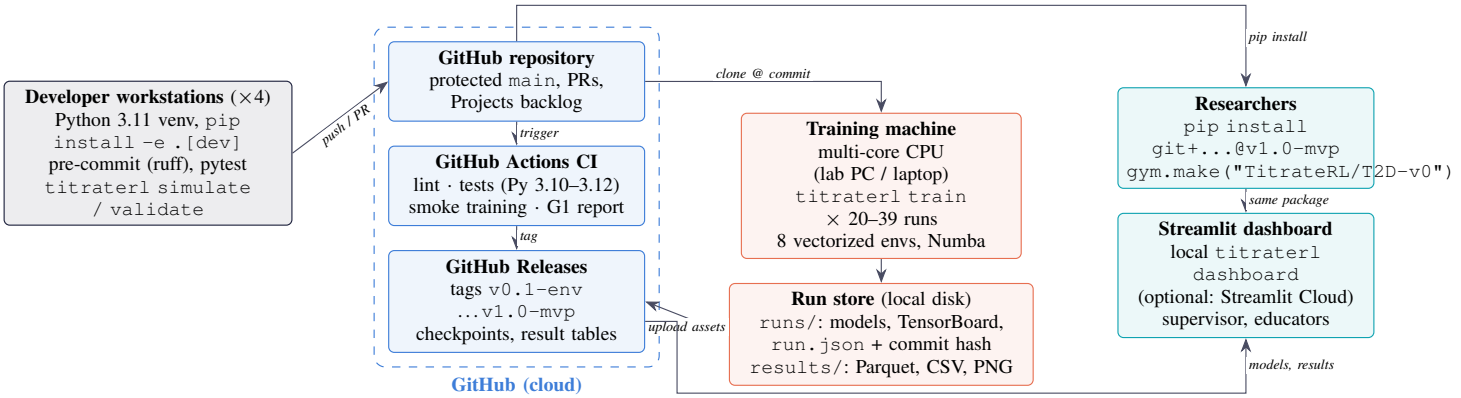


Fig. 1. Deployment map. Code flows from the four workstations through pull requests and CI to tagged releases; experiments run on a multi-core CPU machine from a pinned commit; trained models and result tables are published as release assets (never committed to git) and consumed by the dashboard and by researchers who install the package.

```
titraterl/
├─ .github/ — CI workflow, PR and issue templates
├─ configs/
│  ├─ cohort.yaml
│  ├─ train/ — ppo_r1/r2/r3, ppo_rstar_shield, dqn, fpg, smoke
│  └─ eval/ — test.yaml, baselines_quick.yaml
├─ src/titraterl/
│  ├─ simulation/ — parameters, model (ODE, RK4), scenario, sensors, patient
│  ├─ cohort/ — generator, split, Parquet I/O
│  ├─ envs/ — titration_env, observation, rewards, wrappers
│  ├─ policies/ — base, rules (B0–B4), sb3
│  ├─ training/ — train, callbacks
│  ├─ metrics/ — glycemic, episode
│  ├─ evaluation/ — runner, statistics, logging, pipeline
│  ├─ validation/ — checks V1–V5
│  └─ config.py, cli.py
├─ dashboard/app.py
├─ tests/
│  ├─ unit/ — model, cross-impl., env, shield, rules, ...
│  └─ integration/ — pipeline, CLI, training, dashboard
├─ docs/ — architecture, model, environment, evaluation, ...
├─ examples/ — quickstart.py, quickstart.ipynb
└─ README.md, MODEL_CARD.md, pyproject.toml, Makefile, LICENSE
```

Fig. 2. Repository file structure (86 tracked files, ≈3 800 lines of package code and ≈800 lines of tests).

C. Repository Structure and Class Design

The repository (Fig. 2) follows the `src` layout and mirrors the architecture layers. Fig. 3 shows the main classes: `T2DTitrationEnv` composes a `VirtualPatient`, which owns the ODE state and three random streams (meals, CGM, fingerstick) so that all policies are evaluated with common random numbers. Rule-based baselines and RL agents share one abstract `Policy` interface, and `SafetyShield` is a Gymnasium wrapper applied only to RL agents.

D. Simulation Environment and Virtual Cohort

The patient model is the Dalla Man meal model with published T2D parameters [8], [19]. The glucose, insulin and gut subsystems follow `simglucose` [10] and are extended with β-cell secretion, time-varying hepatic extraction and a two-compartment glargine model [20], [21]. The 15-state system is integrated with a 1-min fourth-order Runge–Kutta scheme compiled with Numba [22]. Each day contains three meals with random timing and size; a CGM samples interstitial glucose every 5 min with the Breton–Kovatchev noise model [23], and a fasting fingerstick is taken at 07:00 [24]. The cohort of 150 virtual adults varies pre-insulin FPG, body weight, insulin sensitivity (ρ_{SI}) and β-cell function (ρ_β), is calibrated against the Treat-to-Target trial [5], and is split 90/30/30 into training, validation and test patients by ρ_{SI} tertile.

Titration is a partially observable decision process exposed through the Gymnasium API [25]: one step is one day and one episode lasts 168 days. The observation stacks 11 daily features (fasting SMBG, dose per kg, last dose change, CGM TIR, TBR, TAR, mean and variability, recent hypoglycemic events, body weight and time) over three days. The action is a dose change in $\{-4, -2, 0, +2, +4\}$ U; the cumulative increase over any seven days is limited to 8 U and the dose to $[0, 1.0$ U/kg$]$ for every policy. The rewards are R1 $=$ TIR, R2 $=$ the negative mean of the Kovatchev risk function [26], and R3 $=$ TIR $- 5\,\mathrm{TBR}_{<70} - 10\,\mathrm{TBR}_{<54}$.

E. Agents, Safety Shield and Baselines

PPO [27] is the primary algorithm and DQN [28] a secondary one, both from Stable-Baselines3 [29]. Each configuration is trained for 10^6 simulated days with five seeds [30], a new training patient in every episode, and checkpoint selection on validation patients only. A safety shield inspired by shielding in safe RL [31] blocks dose increases after level-1 hypoglycemia in the last 72 h and reduces the dose by 20% after level-2 hypoglycemia in the last 24 h. The baselines are a fixed dose of 0.2 U/kg (B0), AT.LANTUS (B1), Treat-to-Target (B2), INSIGHT (B3) and a simulator-tuned rule (B4).

F. Preliminary Implementation Evidence

The environment, baselines, training pipeline, evaluation pipeline and dashboard are implemented. Table II records the simulator checks and a preliminary run of the clinical baselines. Calibration (check V4) does not yet meet all trial tolerances: the virtual patients reach a lower fasting glucose

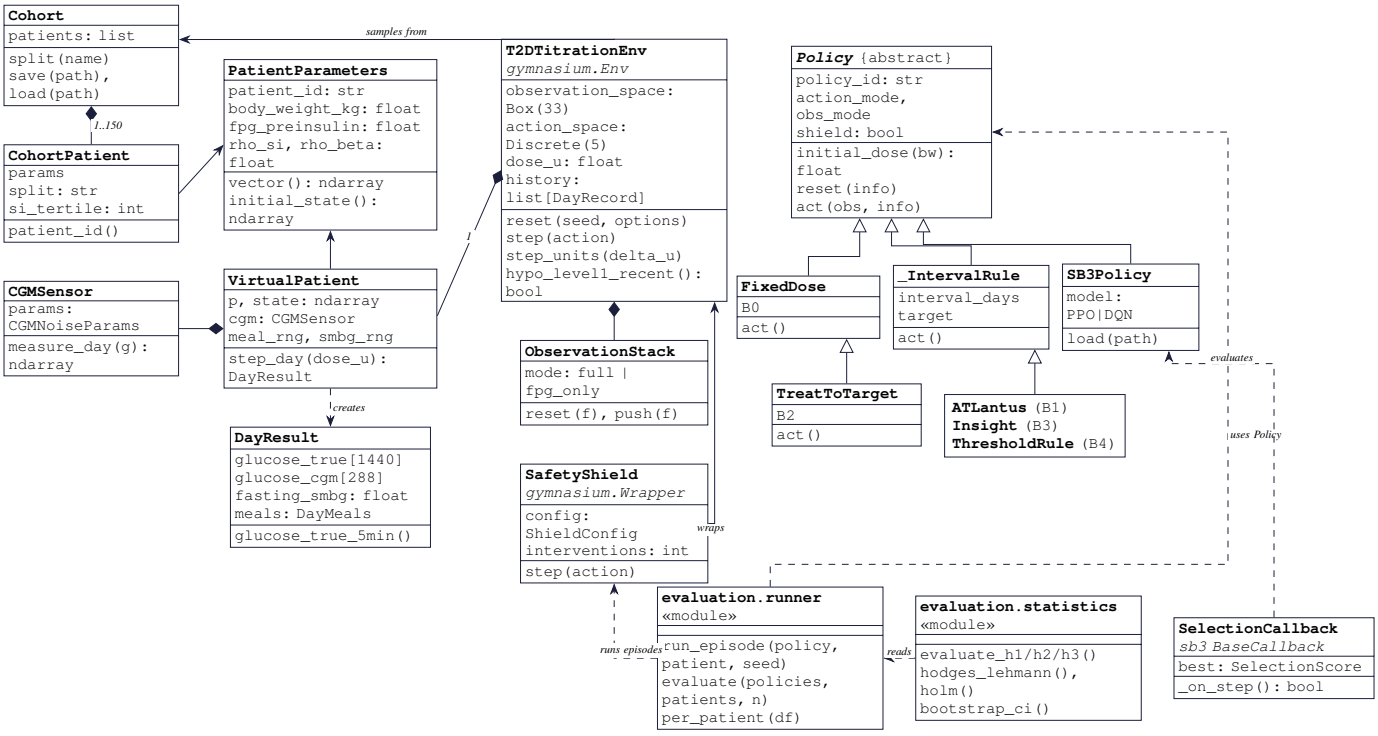


Fig. 3. Class diagram of the implemented TitrateRL package (simplified UML). Open triangles: inheritance; filled diamonds: composition; solid arrows: association; dashed arrows: dependency. B1, B3 and B4 inherit from `_IntervalRule`; B2 from `Policy`.

TABLE II
PRELIMINARY SIMULATOR CHECKS AND BASELINE RUN

Check	Result	Criterion
RK4 vs. SciPy integrator	max error 6×10^{-8} mg/dL	$<$0.5 mg/dL
Right-hand side vs. `simglucose`	identical (10^{-9})	match
Equilibrium; dose–response	drift 3×10^{-14}; monotone	pass
Glargine flatness, 2–24 h	23.7% from mean	$\leq$25%
Glargine steady state, dose 4	90.3%	$\geq$90%
B2 week-24 FPG (90 patients)	99.6 mg/dL	110–125
B2 final dose	0.57 U/kg	0.4–0.6
B2 patients with FPG $\leq$100	56%	25–50%
Runtime, 168-day episode	0.15 s	$<$2 s
Maintenance TIR, B0/B1/B2/B3[†]	39 / 60 / 63 / 61%	—

[†] Validation patients, one scenario each; preliminary software check.

TABLE III
PLANNED MODEL AND SYSTEM EVALUATION METRICS

Metric	Definition	Target / role
Product (acceptance)		
Test suite	unit + integration tests in CI	100% pass
Episode runtime	one 168-day episode	$<$2 s
Dashboard re-run	24-week simulation	$<$5 s
Simulator checks	V1–V4 (gate G1)	all pass by Wk 5
Reproducibility	same seed $\Rightarrow$ same trajectory	exact
Research (glycemic)		
TIR	% time 70–180 mg/dL, weeks 21–24	$>$70%; primary efficacy
$\text{TBR}_{<70}$, $\text{TBR}_{<54}$	% time below 70 / 54 mg/dL	$<$4%, $<$1%; primary safety
$\text{TAR}_{>180}$, $\text{TAR}_{>250}$	% time above 180 / 250	$<$25%, $<$5%
Mean, CV, GMI	mean, SD/mean, $3.31+0.02392\,\bar{G}$	CV $\leq$36%
LBGI, HBGI	Kovatchev risk indices	risk
Time-to-target	3 consecutive fasting values 80–130	titration speed
Nocturnal TBR	00:00–06:00	safety
Overbasalization	bedtime$-$morning $\geq$50; dose $>$0.5 U/kg	appropriateness
Dose burden	final U/kg; changes per week	burden
Shield rate	% days with intervention	safety
Statistical decision rules		
H2 superiority	Holm-adj. one-sided Wilcoxon	$p < 0.05$
H2 effect size	Hodges–Lehmann TIR gain	$\geq$5 pp
H2 non-inferiority	upper 95% CI of $\text{TBR}_{<70}$ diff.	$<$1 pp
H3	Spearman(ρ_{SI}, ΔTIR)	negative, $p < 0.05$

than the trial participants at a comparable final dose. Trained-agent results and held-out evaluation remain planned.

G. Evaluation Protocol

Every policy is evaluated on the 30 test patients, each under ten 168-day scenarios with common random numbers. Endpoints are computed from noise-free glucose following the international CGM consensus [32], [33] (Table III). The primary efficacy endpoint is TIR in weeks 21–24 and the primary safety endpoint is $\text{TBR}_{<70}$ over 24 weeks. H2 is tested with Holm-adjusted one-sided paired Wilcoxon tests, the Hodges–Lehmann estimate of the TIR gain ($\geq$5 pp) and 95% bootstrap confidence intervals (non-inferiority margin 1 pp); RQ3 uses Spearman's correlation. Overbasalization [34] and the glucose management indicator [35] are secondary outcomes.

H. MVP and Implementation Roadmap

The MVP demonstrates the complete pipeline: virtual-patient generation, training and shielded titration, statistical

comparison and interactive visualization. Table IV prioritizes the features with the MoSCoW method, and the team uses Table V to track implementation and evaluation milestones.

TABLE IV
MVP PRIORITIZATION WITH THE MOSCOW METHOD

Priority	Feature	Acceptance criterion	Status
Must	F1 T2D simulator and calibrated 150-patient cohort	generated reproducibly from one seed; checks V1–V4 pass	implemented; V4 open
Must	F2 Gymnasium environment `TitrateRL/T2D-v0`	passes `check_env`; 168-day episode <2 s; rate-limit and shield tests pass	done (0.15 s)
Must	F3 Trained agents: PPO with R1–R3 (shield off) and shielded PPO with $R^\star$, 5 seeds each	checkpoints and learning curves stored; selection on validation patients	pipeline done; runs Wk 7–10
Must	F4 Rule-based policies B0–B3 behind the `Policy` interface	unit tests reproduce the published titration steps	done
Must	F5 Evaluation pipeline	one command regenerates metric tables, statistical tests and figures	done
Must	F6 Interactive dashboard	select a test patient and ≤ 3 policies; traces, doses, metrics, disclaimer; 24-week re-run <5 s; CSV export	done
Must	F7 Installable package	`pip install`; environment registered; documented API and example notebook	done
Must	RQ1–RQ3 analyses	H1–H3 decision rules applied to test-cohort data	Wk 10–13
Should	DQN agent; FPG-only PPO; λ grid for R3; simulator-tuned rule B4; robustness scenario set	completed and reported as exploratory	code ready
Could	Oracle per-patient fine-tuning; 2013 hypoglycemia model extensions; LSTM policy; SAC	started only if M3 is on time	—
Won't (this semester)	Real patient data or clinical deployment; Type 1 diabetes, pumps and bolus insulin; exercise, illness, missed injections; offline RL from EHRs; mobile app	out of scope; stated in README and model card	—

TABLE V
FOURTEEN-WEEK IMPLEMENTATION ROADMAP

Weeks	Planned milestone	Responsible member(s)
1–3	Requirements and architecture; repository, CI and packaging; T2D patient model, glargine model, integrator and cross-implementation tests	N. Khabibullaev, B. Uchqunov, I. Abralov, A. Abdusattarov
4–6	Cohort calibration; Gymnasium environment with reward, rate limit and shield; baselines B0–B3; gate G1; **M1** `v0.1-env`	I. Abralov, A. Abdusattarov, B. Uchqunov, N. Khabibullaev
7–8	PPO training with R1–R3; reward selection and power check (gate G2); **M2** `v0.2-agents`	A. Abdusattarov, N. Khabibullaev, B. Uchqunov
9–10	Evaluation pipeline, statistics and dashboard; shielded PPO-$R^\star$; RQ1 test evaluation	N. Khabibullaev, A. Abdusattarov, I. Abralov, B. Uchqunov
11–12	Full test-cohort evaluation (RQ2); packaging; buffer week; **M3** `v0.3-eval`	N. Khabibullaev, B. Uchqunov, A. Abdusattarov, I. Abralov
13	RQ3 analyses: correlation, dose behavior, generalization gap	N. Khabibullaev, A. Abdusattarov
14	Final report, documentation, model card and demonstration; **M4** `v1.0-mvp`	All members

III. Results

A. Overall Prototype Technical Evaluation (General Objective)

[Insert Results for General Objective here: High-level overview of the integrated titration framework, simulator verification and calibration checks (gate G1), runtime per episode, training throughput, and end-to-end pipeline testing.]

B. Reward Design (SO1)

[Insert Results for Reward Design here: TIR, $\text{TBR}_{<70}$, and $\text{TBR}_{<54}$ of PPO trained with R1, R2, and R3 on the held-out test cohort; median and 95% bootstrap CI; share of patients meeting the consensus targets; H1 test.]

C. Comparison with Clinical Algorithms (SO2)

[Insert Results for Comparison with Clinical Algorithms here: Maintenance-phase TIR and full-period $\text{TBR}_{<70}$ of the shielded PPO policy versus AT.LANTUS, Treat-to-Target, and INSIGHT; Hodges–Lehmann gains, Holm-adjusted p-values, non-inferiority intervals, time-to-target, and dose burden.]

D. Personalization by Insulin Sensitivity (SO3)

[Insert Results for Personalization here: Spearman correlation between insulin sensitivity and the TIR benefit; dose changes at matched fasting glucose by insulin-sensitivity tertile; train–test generalization gap.]

IV. Discussion

A. Overall Prototype Evaluation and Technical Feasibility

[Insert Discussion for General Objective here: Interpretation of simulator validity and calibration deviations, reproducibility of the pipeline, computational feasibility, and in silico scope boundaries.]

B. Reward Design (RQ1 / SO1)

[Insert Discussion for Reward Design here: Interpretation of RQ1, the efficacy–safety trade-off linked to Problem 1, overbasalization risk, and implications for encoding asymmetric clinical objectives.]

C. Comparison with Clinical Algorithms (RQ2 / SO2)

[Insert Discussion for Comparison with Clinical Algorithms here: Interpretation of RQ2 linked to Problem 2, the value of CGM features versus fasting-glucose-only titration, decision frequency, and simulator-specific tuning relative to related work.]

D. Personalization by Insulin Sensitivity (RQ3 / SO3)

[Insert Discussion for Personalization here: Interpretation of RQ3 linked to Problem 3, adaptation to unobserved insulin sensitivity, generalization to unseen patients, and limitations of in silico evidence.]

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